a

Knowledge and forecasting gaps on pre-cutoff events

Knowledge gap · Forecasting gap

Gap value

Feasibility Assessment: GPT-5.4 +0.16 / +0.02; Claude S4.5 +0.20 / +0.04; Claude S5 +0.14 / +0.02; GPT-5.6 Sol +0.08 / +0.02

Mechanistic Forecasting: GPT-5.4 +0.09 / 0.00; Claude S4.5 +0.02 / -0.01; Claude S5 +0.06 / +0.02; GPT-5.6 Sol +0.07 / +0.01

Temporal Prediction: GPT-5.4 +0.45 / +0.02; Claude S4.5 +0.29 / -0.07; Claude S5 +0.33 / +0.02; GPT-5.6 Sol +0.41 / +0.02

Solution Design: GPT-5.4 +0.11 / 0.00; Claude S4.5 +0.08 / +0.01; Claude S5 +0.14 / 0.00; GPT-5.6 Sol +0.11 / +0.01

Knowledge and forecasting gaps on post-cutoff events

Gap value

Feasibility Assessment: GPT-4o +0.07 / +0.11; LLaMA 3.3 +0.04 / +0.04; GPT-5.4 +0.03 / +0.07; Claude S4.5 +0.05 / +0.17; Claude S5 +0.01 / +0.19; GPT-5.6 Sol 0.00 / +0.13

Mechanistic Forecasting: GPT-4o +0.10 / +0.24; LLaMA 3.3 +0.07 / +0.28; GPT-5.4 -0.05 / +0.13; Claude S4.5 0.00 / +0.10; Claude S5 -0.01 / +0.07; GPT-5.6 Sol -0.01 / +0.08

Temporal Prediction: GPT-4o +0.08 / +0.10; LLaMA 3.3 -0.05 / +0.11; GPT-5.4 +0.09 / +0.39; Claude S4.5 -0.04 / +0.18; Claude S5 +0.09 / +0.38; GPT-5.6 Sol +0.09 / +0.38

Solution Design: GPT-4o +0.05 / +0.07; LLaMA 3.3 +0.05 / +0.07; GPT-5.4 +0.04 / +0.07; Claude S4.5 +0.02 / +0.05; Claude S5 +0.04 / +0.08; GPT-5.6 Sol +0.03 / +0.07

b

Feasibility Assessment

Balanced accuracy

Mechanistic Forecasting

Accuracy

Temporal Prediction

Score (0–1)

Solution Design

Score (0–10)

Chemistry · Mat. Sci. · AI · Medicine · Physics · Biology · Neurosci. · Env. Sci.

Cross-model mean · Chance / pass threshold · GPT-4o · GPT-OSS · DeepSeek R1† · LLaMA 3.3 · Claude S4.5 · GPT-5.4 · Claude O4.8 · Claude S5 · GPT-5.6 Sol